Forward and inverse processes in model generation

IA pour la Science

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Two aspects of the SCM inference were discussed the last week:

1. The inference avoids the Integer Liner Programming problem by replacing the probabilities of edges
$$\boldsymbol{\Gamma} \in [0,1]^{n\times n} \mapsto \mathbf{Z} \in \{0,1\}^{n\times n}.$$
for the adjacency of edges. *NB: it is solved below.*

2. The list of variables is not fixed. For the set of observable variables $X_1, \dots, X_n$ the model has to forecast the list of variables $X_1, \dots, X_n, \dots, X_{n'}$. By default, the value $n' = 0$. *NB: it is under consideration.*

The foundation model includes the elements of SCM selection:

1) maximizing likelihood of the DAG at each step,

2) do-intervention actions each step on the sampled node.

The foundation model for the SCM inference

The main type of the foundation model carries the (Riemannian) diffusion process.

The diffusion process involves the step t and two distributions to approximate the distribution of the adjacency matrix $\mathbf{Z}$. Use the vector form $\mathbf{z} = \mathrm{vec}(\mathbf{Z})$. Denote the training distribution $q(\mathbf{z})$ and the sampling distribution $p(\mathbf{z})$. During the model training one has to minimize the distance between two distributions at each step.

The diffusion process invokes the local structural modeling:

$$\mathbf{Z} \mapsto f_g \mapsto \eta \mapsto \mathbf{w},$$

with η is the optimization algorithm that delivers the optimal parameters to the SCM at each step of the diffusion process.

The *fine-tuning* algorithm η optimizes the parameters according to the diagram

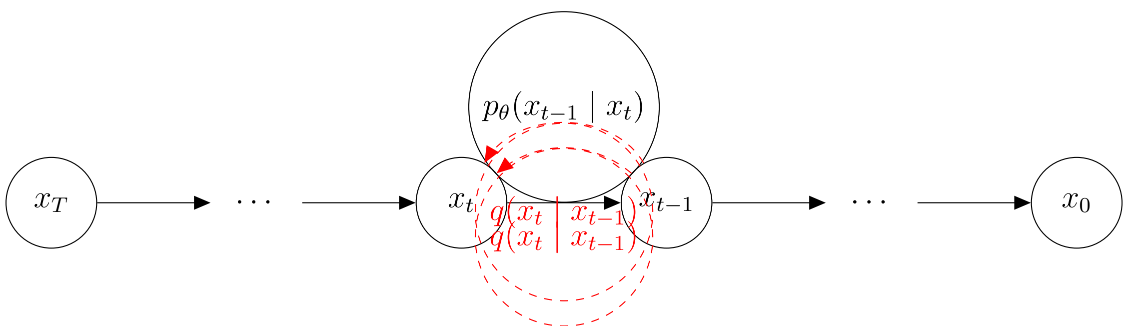
$$
\begin{tikzcd}
\eta(-\lambda^2\nabla_{\mathbf{w}}\mathcal{L}+\mathbf{w}') \arrow[r, "\eta"] & (\hat{\mathbf{w}},\mathbf{x}) \arrow[r, "f_G"] & \hat{\mathbf{x}} \arrow[r] & \mathcal{L}(\mathbf{w}\mid\mathbf{x},\hat{\mathbf{x}}) \arrow[lll, bend right] \\
(t,s) \arrow[r] & i \arrow[r] & \mathbf{x} \arrow[ul] \arrow[ur] &
\end{tikzcd}
$$

This diagram reads: the optimization algorithm η estimates the SCM parameters $\mathbf{w}$ by the gradient of its loss and the previous parameters to solve the problem

$$\hat{\mathbf{w}} = \arg\min \mathcal{L}(\mathbf{w}) \quad \text{given} \quad f_G(\hat{\mathbf{w}},\mathbf{x}) \mapsto \hat{\mathbf{x}}.$$

The algorithm η uses some initial value $\mathbf{w}_0$ and a stop-criterion for its iterations.

A basic diffusion process for the binary adjacent z



Given the data distribution $\mathbf{x}_0 \sim q(\mathbf{x})$ assert:

1) forward diffusion process $\mathbf{x}_t = \sqrt{1-\beta_t}\mathbf{x}_{t-1} + \sqrt{\beta_t}\mathbf{z}_t$, sampling i.i.d.

$$\mathbf{z}_1,\ldots,\mathbf{z}_T \sim \mathcal{N}(\mathbf{0},\mathbf{I}),$$

2) sampling $q(\mathbf{x}_{t-1} \mid \mathbf{x}_t)$ and learning the parameters $\boldsymbol{\theta}$ of f_{ai}: $p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1} \mid \mathbf{x}_t)$,

3) reverse diffusion process

$$p_{\boldsymbol{\theta}}(\mathbf{x}_{0:T}) = p(\mathbf{x}_T)\prod_{t=1}^{T} p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1} \mid \mathbf{x}_t),$$

4) slow learning gives

$$p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1} \mid \mathbf{x}_t) = \mathcal{N}\big(\mathbf{x}_{t-1}; \boldsymbol{\mu}_{\boldsymbol{\theta}}\big(\mathbf{x}_t,t), \boldsymbol{\Sigma}_{\boldsymbol{\theta}}(\mathbf{x}_t,t)\big).$$

The do-intervention and SCM sub-modeling